

Homework — 1.2.1 Operating Systems

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 1.2.1 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Complete the passage about the purpose of an operating system using words from the word bank. Not all words are used. [2] (AO1)

An operating system hides the complexity of the hardware from the user and the programmer — this is known as (a) _____. It also acts as a (b) _____, providing a consistent environment and set of services on which applications software can run. On a single-core CPU the OS creates the illusion that several programs run at once — (c) _____ — by switching rapidly between processes, giving each a short slice of CPU time. Deciding which process gets the CPU next is the job of the (d) _____.

Word bank: abstraction, compression, defragmentation, interrupt, multi-tasking, platform, scheduler

2. The **round robin** row of the table has been completed as an example. Complete the remaining four cells for **first come first served (FCFS)** and **shortest job first (SJF)**. [4] (AO1/AO2)

Scheduling algorithm	How it decides which process runs	One drawback
Round robin (example)	Each process in turn is given a fixed time slice (quantum) of CPU time	Long jobs need many time slices to finish; time is wasted switching between processes
First come first served	(a)	(b)
Shortest job first	(c)	(d)

3. Tick **True** or **False** for each statement. If a statement is false, write a correction in the final column. [3] (AO1)

Statement	True / False	Correction if false
On a single-core processor, multi-tasking is possible because the scheduler switches between processes so quickly that they appear to run simultaneously.		
The BIOS is stored in RAM so that it is available as soon as the computer is switched on.		
Shortest job first can only work if the run time of each job is known (or estimated) in advance.		

4. The steps taken to handle an **interrupt** are shown below in a jumbled order. Number them **1–5** to show the correct order. [2] (AO1)

Step	Order (1–5)
The contents of the registers (e.g. PC, ACC) are pushed onto the stack	
The CPU completes the current fetch–decode–execute cycle and checks the interrupt register	
The registers are popped off the stack and the original process resumes	
The priority of the interrupt is compared with the current task; it is serviced only if it has a higher priority	
The PC is set to the address of the interrupt service routine (ISR), which is then executed	

5. A computer with 8 GB of RAM is running a video editor that needs more memory than is physically available. Explain how **virtual memory** allows the editor to run, and why performance can drop severely (**disk thrashing**) if too many large programs are open at once. [3] (AO2)

Part B — Retrieval: keep it fresh (~6 marks)

6. State the role of the **Program Counter (PC)** and the **Memory Address Register (MAR)** during the fetch stage of the fetch–decode–execute cycle. [2] (1.1.1)

7. Explain how **increasing the number of cores** in a processor can improve performance, and give **one** reason why doubling the cores does not usually double real-world performance. [2] (1.1.1)

8. State **one** advantage of a **solid state drive (SSD)** over a magnetic **hard disk drive (HDD)**, and **one** advantage of an HDD over an SSD. [2] (1.1.3)

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